

Project Development Phase

Preprocessing Steps and Business Questions

Date	15 July 2026
Team ID	
Project Name	Cosmetics Insights
Maximum Marks	

Preprocessing Steps

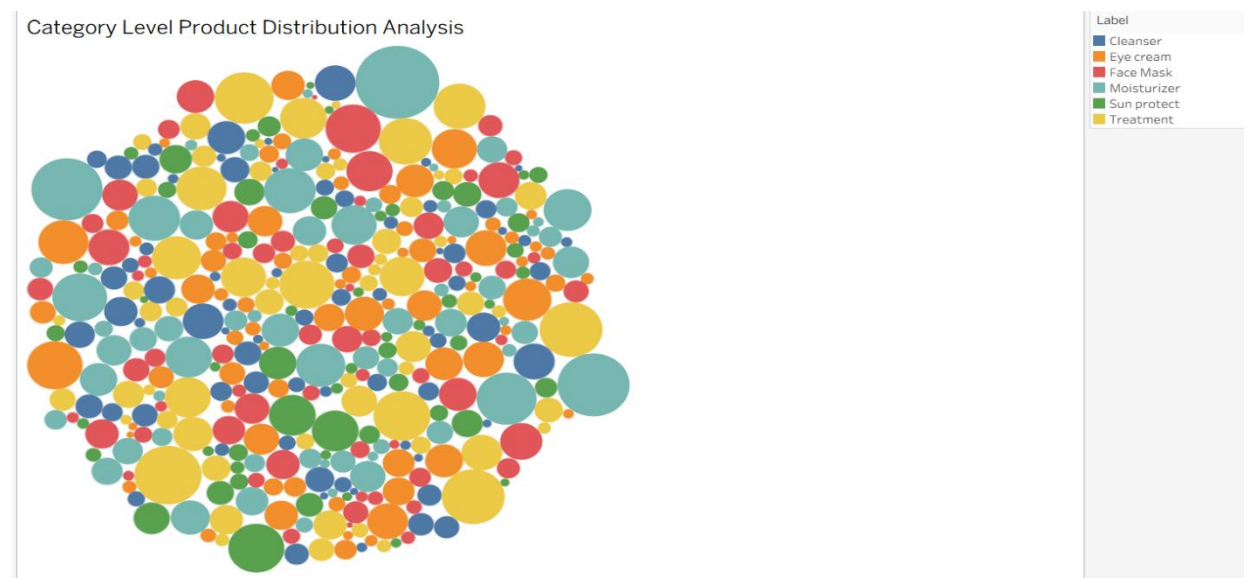
Data preprocessing is an important step that improves the quality of the dataset before visualization. For this project, the cosmetics dataset was carefully cleaned and prepared to ensure accurate analysis and reliable dashboard results. This process helped remove errors, improve consistency, and make the data suitable for creating meaningful visualizations in Tableau.

Preprocessing Steps Performed

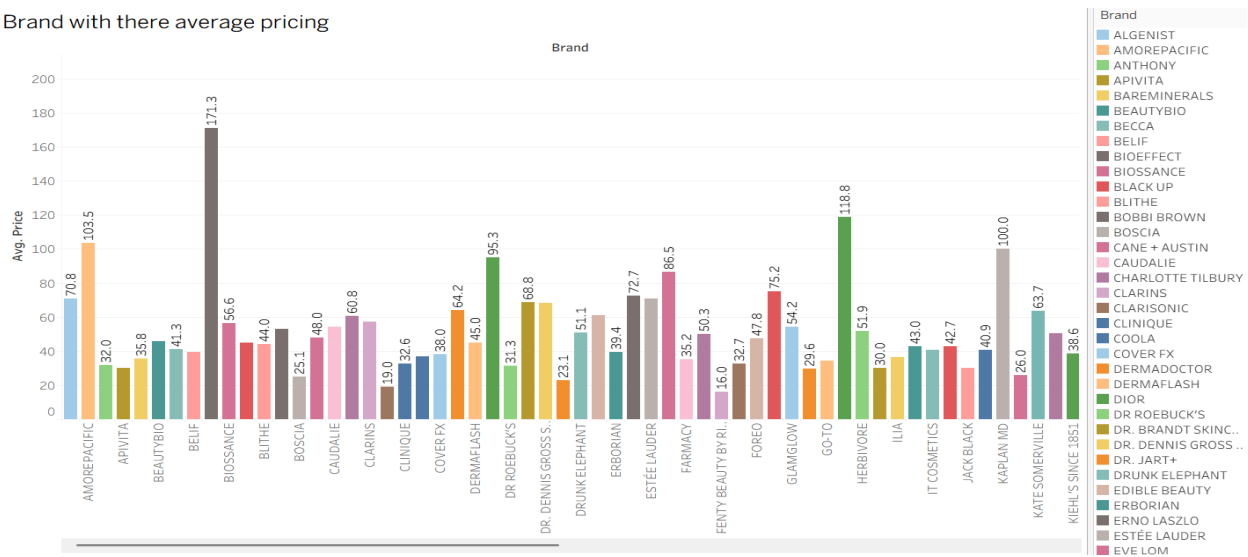
1. Imported the cosmetics dataset into Microsoft Excel.
2. Examined the dataset to understand its structure and identify any data quality issues.
3. Removed duplicate records to avoid repeated information.
4. Checked for missing or incomplete values and handled them appropriately.
5. Corrected inconsistent values and standardized the data format.
6. Renamed column headers to make them clear and easy to understand.
7. Verified the data types for fields such as price, rating, category, and brand.
8. Saved the cleaned dataset and imported it into Tableau Desktop.
9. Created calculated fields, filters, and KPIs to support dashboard analysis.
10. Validated the processed data to ensure accurate visualizations and business insights.

Business Questions with Visualizations

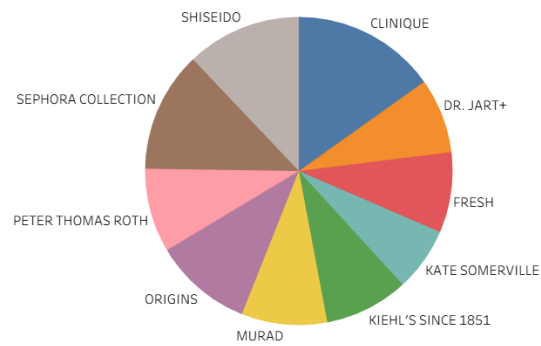
1. Category Level Product Analysis
2. Brand with average pricings
3. Top 10 Brands with Cosmetics Count
4. Category wise Distribution of combination Skin products
5. Category wise total revenue
6. Top 10 brands by product count
7. Price Vs Rank



Brand with there average pricing



Top 10 Brands with cosmetics count



Brand
CLINIQUE
DR. JART+
FRESH
KATE SOMERVILLE
KIEHL'S SINCE 1851
MURAD
ORIGINS
PETER THOMAS ROTH
SEPHORA COLLECTION
SHISEIDO

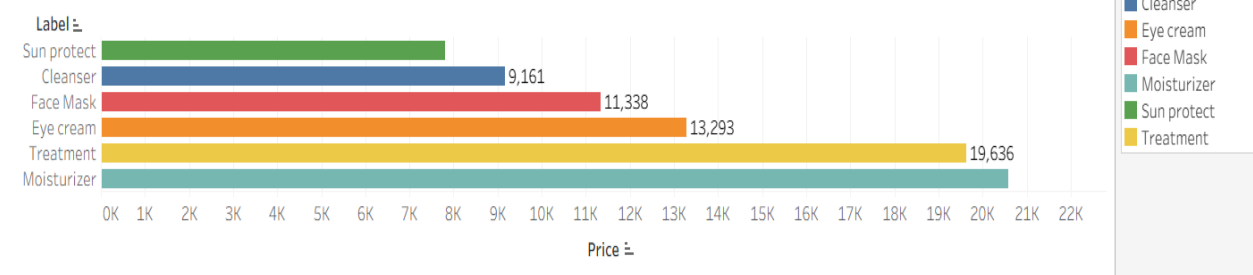
CNT(cosmetics)
521

Category-wise Distribution of Combination Skin Products

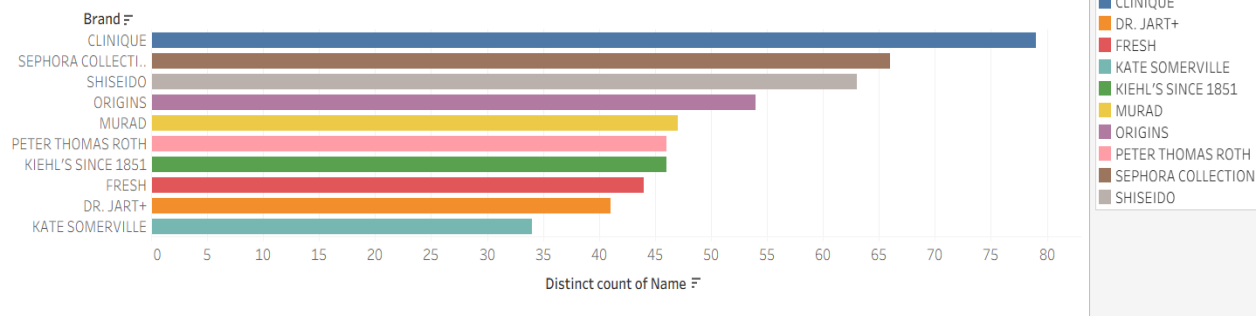


Label
Cleanser
Eye cream
Face Mask
Moisturizer
Sun protect
Treatment

Category wise total revenue



Top 10 brands by product count



Price vs Rank

